

A CONFIRMATORY AUDIT

4 public endpoints

BBBP | ClinTox | ESOL | Lipophilicity

3 label-blind split designs

Random | scaffold | similarity cluster

5 reliability lenses

Performance | calibration | domain
conformal sets | selective retention

Paired, frozen, repeated-seed evidence

B FAILURE SIGNALS

0.90 overall

0.07-0.10 positive
ClinTox marginal coverage

~71% ambiguous

Mondrian repairs coverage
by returning {0,1}

27-31% retained

ClinTox positives when
50% of samples remain

C TAME DESIGN

Two label-free transport views

Physicochemical descriptors + model scores

Transport audit

Baseline-containing consensus

Expand uncertainty; never invent confidence

Audit failure: ordinary baseline fallback

D SEALED EPA TEST

predictions

HASH + SEAL

then open labels

+1.36 pp

minimum-class coverage

95% CI +0.58 to +2.01 pp

MacroCSY -1.61 pp

within the frozen -5 pp non-inferiority bound

60/60 endpoint-seed cells complete

DIAGNOSIS > DESIGN CONSTRAINTS > AUDITABLE METHOD > INDEPENDENT EVIDENCE